complex tasks, showcasing advanced communication capabilities, thereby realizing self-driven evolution in their interactions with the environment. In CLMTWA [99], by using a large language model as a teacher and a weaker language model as a student, the teacher can generate and communicate natural language explanations to improve the student's reasoning skills via theory of mind. The teacher can also personalize its explanations for the student and intervene only when necessary, based on the expected utility of intervention. Meanwhile, NLSOM [100],leverages natural language collaboration between agents, dynamically adjusting roles, tasks, and relationships based on feedback to solve problems beyond the scope of a single agent.

Remark. Upon comparing the aforementioned strategies for agent capability acquisition, we can find that the fine-tuning method improves the agent capability by adjusting model parameters, which can incorporate a large amount of task-specific knowledge, but is only suitable for open-source LLMs. The method without fine-tuning usually enhances the agent capability based on delicate prompting strategies or mechanism engineering. They can be used for both open- and closed-source LLMs. However, due to the limitation of the input context window of LLMs, they cannot incorporate too much task information. In addition, the designing spaces of the prompts and mechanisms are extremely large, which makes it not easy to find optimal solutions.

In the above sections, we have detailed the construction of LLM-based agents, where we focus on two aspects including the architecture design and capability acquisition. We present the correspondence between existing work and the above taxonomy in Table 1. It should be noted that, for the sake of integrity, we have also incorporated several studies, which do not explicitly mention LLM-based agents but are highly related to this area.

3 LLM-based Autonomous Agent Application

Owing to the strong language comprehension, complex task reasoning, and common sense understanding capabilities, LLM-based autonomous agents have shown significant potential to influence multiple domains. This section provides a succinct summary of previous studies, categorizing them according to their applications in three distinct areas: social science, natural science, and engineering (see the left part of Figure 5 for a global overview).

3.1 Social Science

Social science is one of the branches of science, devoted to the study of societies and the relationships among individuals within those societies. LLM-based autonomous agents can promote this domain by leveraging their impressive human-like understanding, thinking and task solving capabilities. In the following, we discuss several key areas that can be affected by LLM-based autonomous agents.

Psychology: For the domain of psychology, LLM-based agents can be leveraged for conducting simulation experiments, providing mental health support and so on [101–104]. For example, in [101], the authors assign LLMs with different profiles, and let them complete psychology experiments. From the results, the authors find that LLMs are capable of generating results that align with those from studies involving human participants. Additionally, it was observed that larger models tend to deliver more accurate simulation results compared to their smaller counterparts. An interesting discovery is that, in many experiments, models like ChatGPT and GPT-4 can provide too perfect estimates (called

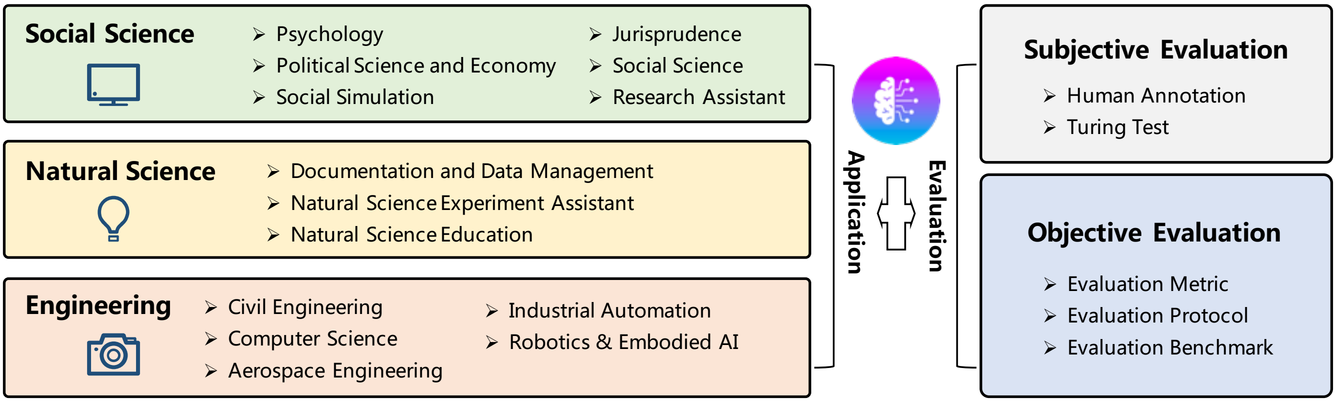

Fig. 5 The applications (left) and evaluation strategies (right) of LLM-based agents.

"hyper-accuracy distortion"), which may influence the downstream applications. In [103], the authors systematically analyze the effectiveness of LLM-based conversation agents for mental well-being support. They collect 120 posts from Reddit, and find that such agents can help users cope with anxieties, social isolation and depression on demand. At the same time, they also find that the agents may produce harmful contents sometimes.

Political Science and Economy: LLM-based agents can also be utilized to study political science and economy [29, 104, 105]. In [29], LLM-based agents are utilized for ideology detection and predicting voting patterns. In [104], the authors focuses on understanding the discourse structure and persuasive elements of political speech through the assistance of LLM-based agents. In [105], LLM-based agents are provided with specific traits such as talents, preferences, and personalities to explore human economic behaviors in simulated scenarios.

Social Simulation: Previously, conducting experiments with human societies is often expensive, unethical, or even infeasible. With the ever prospering of LLMs, many people explore to build virtual environment with LLM-based agents to simulate social phenomena, such as the propagation of harmful information, and so on [20, 34, 78, 80, 106–109]. For example, Social Simulacra [80] simulates an online social community and explores the potential of utilizing agent-based simulations to aid decision-makers to improve community regulations. [106, 107] investigates the potential impacts of different behavioral characteristics of LLM-based agents in social networks. Generative Agents [20] and AgentSims [34] construct multiple agents in a virtual town to simulate the human daily life. SocialAI School [108] employs LLM-based agents to simulate and investigate the fundamental social cognitive skills during the course of child development. S^3 [78] builds a social network simulator, focusing on the propagation of information, emotion and attitude. CGMI [110] is a framework for multi-agent simulation. CGMI maintains the personality of the agents through a tree structure and constructs a cognitive model. The authors simulated a classroom scenario using CGMI.

Jurisprudence: LLM-based agents can serve as aids in legal decision-making processes, facilitating more informed judgements [111, 112]. Blind Judgement [112] employs several language models to simulate the decision-making processes of multiple judges. It gathers diverse opinions and consolidates the outcomes through a voting mechanism. ChatLaw [111] is a prominent Chinese legal model based on LLM. It adeptly supports both database and keyword search strategies, specifically designed to mitigate the hallucination issue prevalent in such models. In addition, this model also employs self-attention mechanism to enhance the LLM's capability via mitigating the impact of reference inaccuracies.

Research Assistant: Beyond their application in specialized domains, LLM-based agents are increasingly adopted as versatile assistants in the broad field of social science research [104, 113]. In [104], LLM-based agents offer multifaceted assistance, ranging from generating concise article abstracts and extracting pivotal keywords to crafting detailed scripts for studies, showcasing their ability to enrich and streamline the research process. Meanwhile, in [113], LLM-based agents serve as a writing assistant, demonstrating their capability to identify novel research inquiries for social scientists, thereby opening new avenues for exploration and innovation in the field. These examples highlight the potential of LLM-based agents in enhancing the efficiency, creativity, and breadth of social science research.

3.2 Natural Science

Natural science is one of the branches of science concerned with the description, understanding and prediction of natural phenomena, based on empirical evidence from observation and experimentation. With the ever prospering of LLMs, the application of LLM-based agents in natural sciences becomes more and more popular. In the following, we present many representative areas, where LLM-based agents can play important roles.

Documentation and Data Management: Natural scientific research often involves the collection, organization, and synthesis of substantial amounts of literature, which requires a significant dedication of time and human resources. LLM-based agents have shown strong capabilities on language understanding and employing tools such as the internet and databases for text processing. These capabilities empower the agent to excel in tasks related to documentation and data management. In [114], the agent can efficiently query and utilize internet information to complete tasks such as question answering and experiment planning. ChatMOF [115] utilizes LLMs to extract important information from text descriptions written by humans. It then formulates a plan to apply relevant tools for predicting the properties and structures of metal-organic frameworks. ChemCrow [76] utilizes chemistry-related databases to both validate the precision of compound representations and identify potentially dangerous substances. This functionality enhances the reliability and comprehensiveness of scientific inquiries by ensuring the accuracy of the data involved.

Experiment Assistant: LLM-based agents have the ability to independently conduct experiments, making them valuable tools for supporting scientists in their research projects [76,114]. For instance, [114] introduces an innovative agent system that utilizes LLMs for automating the design, planning, and execution of scientific experiments. This system, when provided with the experimental objectives as input, accesses the Internet and retrieves relevant documents to gather the necessary information. It subsequently utilizes Python code to conduct essential calculations and carry out the following experiments. ChemCrow [76] incorporates 17 carefully developed tools that are specifically designed to assist researchers in their chemical research. Once the input objectives are received, ChemCrow provides valuable recommendations for experimental procedures, while also emphasizing any potential safety risks associated with the proposed experiments.

Natural Science Education: LLM-based agents can communicate with humans fluently, often being utilized to develop agent-based educational tools. For example, [114] develops agent-based education systems to facilitate students learning of experimental design, methodologies, and analysis. The objective of these systems is to enhance students' critical thinking and problem-solving skills, while also fostering a deeper comprehension of scientific principles. Math Agents [116] can assist researchers in exploring, discovering, solving and proving mathematical problems. Additionally, it can communicate with humans and aids them in understanding and using mathematics. [117] utilize the capabilities of CodeX [118] to automatically solve and explain university-level mathematical problems, which can be used as education tools to teach students and researchers. CodeHelp [119] is an education agent for programming. It offers many useful features, such as setting course-specific keywords, monitoring student queries, and providing feedback to the system. EduChat [87] is an LLM-based agent designed specifically for the education domain. It provides personalized, equitable, and empathetic educational support to teachers, students, and parents through dialogue. FreeText [120] is an agent that utilizes LLMs to automatically assess students' responses to open-ended questions and offer feedback.

3.3 Engineering

LLM-based autonomous agents have shown great potential in assisting and enhancing engineering research and applications. In this section, we review and summarize the applications of LLM-based agents in several major engineering domains.

Computer Science & Software Engineering: In the field of computer science and software engineering, LLM-based agents offer potential for automating coding, testing, debugging, and documentation generation [18, 23, 24, 126–128]. ChatDev [18] proposes an end-to-end framework, where multiple agent roles communicate and collaborate through natural language conversations to complete the software development life cycle. This framework demonstrates efficient and cost-effective generation of executable software systems. MetaGPT [23] abstracts multiple roles, such as product managers, architects, project managers, and engineers, to supervise code generation process and enhance the quality of the final output code. This enables low-cost software development. [24] presents a self-collaboration framework for code generation using LLMs. In this framework, multiple LLMs are assumed to be distinct "experts" for specific subtasks. They collaborate and interact according to specified instructions, forming a virtual team that facilitates each other's work. Ultimately, the virtual team collaboratively addresses code generation tasks without requiring human intervention. LLIFT [141] employs LLMs to assist in conducting static analysis, specifically for identifying potential code vulnerabilities. This approach effectively manages the trade-off between accuracy and scalability. ChatEDA [123] is an agent developed for electronic design automation (EDA) to streamline the design process by integrating task planning, script generation, and execution. CodeHelp [119] is an agent designed to assist students and developers in debugging and testing their code. Its features include providing detailed explanations of error messages, suggesting potential fixes, and ensuring the accuracy of the code. Pentest [125] is a penetration testing tool based on LLMs, which can effectively identify common vulnerabilities, and interpret source code to develop exploits. D-Bot [122] utilizes the capa-

Table 2 Representative applications of LLM-based autonomous agents.

	Domain	Work
Social Science	Psychology	TE [101], Akata et al. [102], Ziems et al. [104], Ma et al. [103]
	Political Science and Economy	Argyle et al. [29], Horton [105], Ziems et al. [104]
	Social Simulation	Social Simulacra [80], Generative Agents [20], SocialAI School [108], AgentSims [34], S^3 [78], Williams et al. [109], Li et al. [106], Chao et al. [107]
	Jurisprudence	ChatLaw [111], Blind Judgement [112]
	Research Assistant	Ziems et al. [104], Bail et al. [113]
Natural Science	Documentation and Data Management	ChemCrow [76], ChatMOF [115], Boiko et al. [114]
	Experiment Assistant	ChemCrow [76], Boiko et al. [114], Grossmann et al. [121]
	Natural Science Education	ChemCrow [76], CodeHelp [119], Boiko et al. [114], MathAgent [116], Drori et al. [117], EduChat [87], FreeText [120]
Engineering	CS & SE	RestGPT [71], Self-collaboration [24], SQL-PALM [89], RAH [91], D-Bot [122], RecMind [53], ChatEDA [123], InteRecAgent [124], PentestGPT [125], CodeHelp [119], SmolModels [126], DemoGPT [127], GPTEngineer [128]
	Industrial Automation	GPT4IA [129], IELLM [130]
	Robotics & Embodied AI	ProAgent [131], LLM4RL [132], PET [133], REMEMBERER [134], DEPS [33], Unified Agent [135], SayCan [79], TidyBot [136], RoCo [92], SayPlan [31], TaPA [137], Dasgupta et al. [138], DECKARD [139], Dialogue shaping [140]

bilities of LLMs to systematically assess potential root causes of anomalies in databases. Through the implementation of a tree of thought approach, D-Bot enables LLMs to backtrack to previous steps in case the current step proves unsuccessful, thus enhancing the accuracy of the diagnosis process.

Industrial Automation: In the field of industrial automation, LLM-based agents can be used to achieve intelligent planning and control of production processes. [129] proposes a novel framework that integrates LLMs with digital twin systems to accommodate flexible production needs. The framework leverages prompt engineering techniques to create LLM agents that can adapt to specific tasks based on the information provided by digital twins. These agents can coordinate a series of atomic functionalities and skills to complete production tasks at different levels. This research demonstrates the potential of integrating LLMs into industrial automation systems, providing innovative solutions for more agile, flexible and adaptive production processes. IELLM [130] showcases a case study on LLMs' role in the oil and gas industry, covering applications like factory automation and PLC programming.

Robotics & Embodied Artificial Intelligence: Recent works have advanced the development of more efficient reinforcement learning agents for